



Synthesis gas conversion over a Rh–K–MoP/SiO₂ catalyst

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ABSTRACT

Synthesis gas conversion over a 1 wt% Rh–5 wt% K–10 wt% MoP on SiO₂ catalyst was investigated at different temperatures, synthesis gas H₂/CO ratios and space velocities. Compared to a 5 wt% K–10 wt% MoP on SiO₂ catalyst, addition of Rh increased the stability of the catalyst and the selectivity to hydrocarbons. The highest C₂₊ oxygenate selectivity of 44 C atom% was achieved at 598 K and H₂:CO = 1. Power law kinetics were used to describe the space–time yield (STY) of the major products. The apparent activation energies for ethanol (77.7 kJ/mol) and acetaldehyde (81.9 kJ/mol) formation suggested that both originated from the same surface intermediate whereas, a higher activation barrier was identified for methanol (114.5 kJ/mol).

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1. Introduction

Interest in the development of new technologies that convert renewable resources to alternative transportation fuels, and thereby address issues of air quality, CO₂ emissions and energy security, are increasing. Biomass, especially agricultural and forest residue, has potential as a renewable energy resource and is expected to play an important role in the synthesis of clean and sustainable fuels [1,2]. One route to clean fuels from biomass is through the thermocatalytic conversion of synthesis gas (CO + CO₂ + H₂) that is produced by biomass gasification. The synthesis of alcohols from synthesis gas (syngas) has been known since the 1930s [3,4] and the production of methanol is practiced industrially using Cu/ZnO catalysts [5]. Higher alcohols, especially iso-butanol, are produced when high temperature Zn/Cr and low temperature Cu/ZnO methanol catalysts are promoted with alkali metals. These catalysts produce C₁–C₆ linear and branched alcohols [6,7] primarily via an aldol condensation or methanol homologation reaction mechanism [1,8]. However, they have very low selectivity toward ethanol because of the fast C–C chain growth of the C₂ surface intermediate [8,9].

Many other catalysts for the synthesis of ethanol or mixed alcohols have been reported in the literature, and these can be categorized as (i) modified Fischer–Tropsch (FT) catalysts (ii) Rh-based catalysts and (iii) Mo-based catalysts. Alkali-doped FT catalysts produce mainly hydrocarbons from syngas, with a ratio of hydrocarbon to alcohol equal to one or more [10], and among the alcohols, C₂₊ alcohols predominate. A high yield of alcohols has also been reported on Cu–Co catalysts, but these catalysts produce significant amounts of hydrocarbons which have a negative effect on the overall process feasibility, especially if the CH₄ selectivity is above about 10% [1]. Rh-based catalysts have the highest reported ethanol selectivity among the various synthesis gas catalysts investigated to date [1,11–16]. Hu et al. reported 44.5 C atom% selectivity to ethanol with a considerable amount of CH₄ (48 C atom%) in the product [11] when synthesis gas was reacted over a Rh catalyst, promoted with Li and Mn, at 573 K, 3 MPa and a syngas H₂:CO ratio of 2. Recently, Rh supported on carbon nanotubes was reported to have good selectivity (52.4 C atom%) towards C₂₊ oxygenates as well [12].

The high hydrocarbon selectivity reported on Rh-based catalysts, especially toward CH₄, means that these catalysts would be difficult to implement in a commercial process. Furthermore, because of cost and supply issues associated with Rh, there is significant interest in less expensive synthesis gas conversion catalysts with high alcohol selectivity. Several researchers have reported that Mo sulphides, phosphides, nitrides and carbides have catalytic

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properties similar to some precious metal [17], and that they resist sulphur poisoning [1]. These two characteristics have led to several studies of Mo-based catalysts for syngas conversion to alcohols. Synthesis gas conversion at 568 K, 7.2 MPa and a $H_2:CO$ ratio of 1 was reported with high selectivity to ethanol (40 C atom%) over a K-promoted MoS_2 [18,19], but later researchers only achieved 10–30 C atom% ethanol selectivity on K- MoS_2 catalysts promoted with Rh, Co, Ni, and Mn [20–23]. Recently, syngas conversion over β - Mo_2C at 573 K and 8 MPa was reported to produce hydrocarbons and alcohols, and among the liquid products, ethanol predominated [24,25]. On alkali doped Mo catalysts (both Mo_2C and MoS_2), the higher alcohol synthesis followed the Anderson–Schulz–Flory distribution, with a high selectivity to methanol. Addition of FT metals (Ni, Co, Rh) increased the selectivity to ethanol and enhanced the performance of MoS_2 and Mo_2C catalysts [20–23,25].

The kinetics of synthesis gas conversion to alcohols has also been reported in the literature. The activation energy of synthesis gas conversion on Rh catalysts promoted with Mn has been reported by several researchers [13–16], with values of 90–140 kJ/mol reported for the activation energy of methane formation and 40–70 kJ/mol for ethanol formation. Kinetic models for higher alcohol synthesis over MoS_2 have also been developed by several researchers [26–28] and it is now well established that the alcohol chain growth occurs via a CO insertion mechanism on MoS_2 catalysts, although recent theoretical studies have shown that CO does not dissociate on MoS_2 [29]. There is very limited information on the kinetics of synthesis gas conversion over Mo_2C catalysts, although Xiang et al. [25] has reported the activation energies for linear alcohols (C_1 – C_5) over several alkali-promoted Mo_2C catalysts. For K- Mo_2C the ethanol activation energy was 74 kJ/mol, and with the addition of Co, the activation energy decreased to 71 kJ/mol [25].

Recently, the present authors reported on the product distribution obtained from synthesis gas conversion over a new series of K-MoP catalysts supported on SiO_2 [30,31]. Within the investigated range of compositions, a 5 wt% K–10 wt% MoP supported on SiO_2 catalyst showed the highest selectivity towards liquid oxygenates and C_2 oxygenates, i.e. ethanol (16 C atom%) and acetaldehyde (18 C atom%). Although the MoP catalysts showed promising selectivity to oxygenates and ethanol, the catalysts deactivated during the first 20 h of operation before stabilizing at a lower level of conversion. In the present study, we demonstrate that addition of 1 wt% Rh to the 5 wt% K–10 wt% MoP- SiO_2 catalyst stabilizes the CO conversion. The space-time-yield data for the Rh-K-MoP/ SiO_2 catalyst, measured at similar CO conversions (10–25%), have been used to develop simplified power law kinetic models of the synthesis of the oxygenated compounds and the apparent activation energies obtained for the Rh-K-MoP/ SiO_2 catalyst have been compared to those obtained over other Mo catalysts.

2. Experimental

2.1. Catalyst preparation

The silica supported Rh-K-MoP catalyst was prepared by step-wise impregnation of a SiO_2 support, followed by calcination and temperature-programmed reduction (TPR). Approximately 10 g of the SiO_2 (Sigma-Aldrich, Grade 62, 60–200 mesh, BET area = 330 m²/g, pore volume = 1.2 cm³/g) was impregnated with a solution of potassium nitrate (1.77 g KNO_3 , BDH Chemicals, 99.97% dissolved in 15 ml of de-ionized water). After aging at room temperature for 12 h, the impregnated SiO_2 was dried at 373 K for 12 h followed by calcination at 773 K for 5 h. Stoichiometric amounts (Mo:P = 1) of ammonium heptamolybdate (1.39 g of $(NH_4)_6Mo_7O_{24} \cdot 4H_2O$, BDH Chemicals, 99%) and diammonium hydrogen phos-

phate (1.04 g of $(NH_4)_2HPO_4$, Sigma-Aldrich, 99%) were dissolved in 15.7 ml of de-ionized water and impregnated drop-wise onto 10 g of the K- SiO_2 with continuous mixing. The impregnated support was held at room temperature for 12 h before being dried at 373 K for 12 h and calcined at 773 K for 5 h. To add the 1 wt% Rh, 0.215 g of rhodium (III) acetate (Sigma-Aldrich, 99%) was dissolved in 15 ml of de-ionized water. A few drops of HNO_3 were added to make a translucent solution (light brown-yellow in color). The Rh solution was then added to the calcined K-MoP- SiO_2 , aged for 12 h at room temperature and dried at 373 K for 12 h. A final calcination was conducted at 773 K for 5 h. The calcined catalyst precursor was subjected to TPR in a H_2 (Praxair, 99.99%) flow of 120 cm³ (STP) min⁻¹ g⁻¹, at a temperature ramp of 1 K/min to a final temperature of 923 K. The final temperature was maintained for 2 h. After reduction, the catalyst was cooled in He to the desired reaction temperature. For samples prepared for EDX, XPS and BET area analysis, the catalyst was cooled to room temperature in He and then passivated in 2 vol% O_2 in He for 2 h at room temperature prior to removal from the reactor.

Details of the preparation of the MoP/ SiO_2 and K-MoP/ SiO_2 catalysts used herein for comparison, have been reported previously [30,31].

2.2. Catalyst characterization

The chemical composition of the prepared catalysts was done by CANTEST Laboratories (Burnaby, BC) using ICP-AES. Prior to analysis about 0.1 g of the passivated catalyst was mixed with 0.7 g of $LiBO_2$ and fused at 1273 K before being dissolved in 100 ml of 4% nitric acid.

The catalyst single point BET surface areas were measured using a Micromeritics FlowSorbII 2300 analyser. About 0.1 g of the passivated catalyst was degassed at 473 K for 2 h and the measurement was made using 30% N_2 and 70% He. Temperature-programmed reduction experiments were carried out using a Micromeritics AutoChem 2920 apparatus, following the procedure of Zuzaniuk and Prins [33]. The calcined, catalyst precursor (160–180 mg) was placed in a quartz U-tube and reduced in a flow of a 9.5% H_2 /Ar mixture (50 ml min⁻¹). The temperature was increased at a ramp rate of 5 K min⁻¹ up to 1023 K, the final temperature being maintained for 4 h. Prior to the analysis, the temperature-programmed reduction of a reference material (silver oxide) was carried out using the same procedure. The peak area of the reference material was correlated to the known volume of consumed H_2 and these data were used to calculate H_2 consumption during the reduction of the phosphide precursors.

EDX analysis was performed using a Hitachi S-3000N electron microscope operated with a 20 kV electron beam acceleration voltage. The average composition, from at least 10 data points, was determined for each catalyst sample. A Leybold Max200 X-ray photoelectron spectrometer with an Al $K\alpha$ photon source was used for the XPS analysis. After reduction and reaction the catalyst was cooled to room temperature in He and prior to removal from the reactor, the catalyst was passivated in 2 vol% O_2 in He for 2 h at room temperature. Exposure of the samples to ambient atmosphere was minimized by transferring the samples from the reactor to the spectrometer either in vacuum or under N_2 . No further treatment of the catalysts was done in the XPS chamber. All XPS spectra were corrected to the C1s peak with a binding energy (BE) of 284.6 eV.

2.3. Catalyst assessment

Catalyst activities were measured in a laboratory fixed-bed micro-reactor (copper lined stainless steel tube with o.d. = 9.53 mm and i.d. = 6.35 mm), operated at low conversion to investigate the

Table 1
Comparison of MoP catalyst compositions.

	Nominal (fresh catalyst), wt%	Measured – EDX (fresh catalyst), wt%	Measured – ICP-AES (after reaction), wt%
MoP/SiO ₂			
MoP	10	7.80 ^a	8.30 ^a
SiO ₂	90	92.70	91.70
Mo:P	1.00	0.70	1.02
K–MoP/SiO ₂			
K ₂ O	5	5.80	5.5
MoP	10	8.50 ^a	10.6 ^a
SiO ₂	85	85.60	83.9
Mo:P	1.00	0.72	1.10
Rh–K–MoP/SiO ₂			
Rh	1	1.12	0.85
K ₂ O	5	4.45	5.00
MoP	10	9.15 ^a	9.40 ^a
SiO ₂	84	85.28	84.80
Mo:P	1.00	0.76	1.04

^a Based on the Mo analysis.

kinetics of the reaction. This was achieved by adjusting the gas hourly space velocity (GHSV) at each reaction temperature. Accordingly, at 573 K, 598 K and 613 K, the GHSV was set at 3960, 7920 and 15,840 h^{−1}, by using 1, 0.5 and 0.25 g catalyst, respectively. The calcined catalyst particles (average diameter = 150 μm) were placed on a packed bed of quartz wool inside the reactor and held in place by a bed of quartz beads. Prior to the start of the reaction the calcined catalyst was reduced *in situ* in 120 (STP) cm³ min^{−1} g^{−1} of H₂ at a temperature ramp rate of 1 K/min to a final temperature of 923 K that was held for 2 h before cooling the reactor to the desired reaction temperature in He. The catalyst bed depth was 2–7 cm and calculation showed that this configuration ensured plug-flow through the reactor. A high temperature back pressure regulator was used to control the reactor pressure and a mass flow controller was used to control the flow of pre-mixed gas to the reactor. Various H₂:CO gas mixtures were investigated. The reaction products were analyzed using an in-line gas chromatograph (GC). Light gases (CO, CO₂ and C₁–C₄ hydrocarbons) were separated using a 5 m temperature-programmed Porapak Q 80/100 packed column and quantified with a thermal conductivity detector. The alcohols, aldehydes, ketones, carboxylic acids and C₅ + hydrocarbons were separated using a 30 m temperature-programmed ECTM-wax capillary column (i.d. = 0.53 mm and film thickness 1.20 μm) and quantified using a flame ionization detector. GC/MS analysis was also carried out periodically to confirm the identity of the reaction products.

The activities of the catalysts were measured at different temperatures and GHSVs and at each condition the catalysts were evaluated for a period of at least 60 h of continuous operation. Some experiments were repeated to quantify the experimental error and analysis showed the conversion and selectivity data to be within ± 10% of the reported values. Calculation of the internal and external heat and mass transfer rates, and application of the Mears and Weisz–Prater criteria, confirmed that at the conditions of the experiments, the reactor operated free of significant heat and mass transfer resistances. The carbon balance across the reactor was >98% for all the experimental data reported herein.

3. Results

3.1. Catalyst characterization

Table 1 reports the compositions of the MoP/SiO₂, the K–MoP/SiO₂ and the Rh–K–MoP/SiO₂ catalysts used in the present study. Chemical analysis (ICP-AES) of the catalysts after reaction was in good agreement with the nominal catalyst compositions.

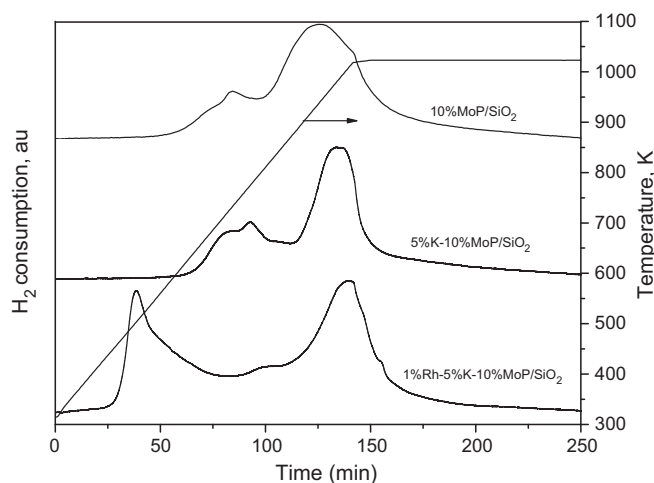


Fig. 1. TPR profile of MoP/SiO₂, K–MoP/SiO₂ and Rh–K–MoP/SiO₂ catalyst.

After reaction, the catalysts were marginally enriched in Mo, possibly a consequence of some P loss during temperature programmed reduction of the calcined catalyst precursors, as reported by Clark and Oyama [32]. Table 1 also shows that the catalyst compositions, as determined by EDX, were in reasonable agreement with the ICP-AES chemical analysis, except that the P content was over estimated by EDX (Mo:P ratio < 1 in all cases) because of poor resolution of the Si and P peaks.

The TPR profile of the calcined Rh–K–MoP/SiO₂ catalyst precursor is compared to that obtained from the MoP/SiO₂ and K–MoP/SiO₂ catalyst precursors in Fig. 1. After calcination, it was assumed that the Mo and P of the catalyst precursors were completely oxidized and present on the support as Mo⁶⁺ and P⁵⁺, respectively. The TPR profile of the 10 wt% MoP–SiO₂ catalyst was in agreement with that reported by Zuzaniuk and Prins [33] with two peak maxima at 727 and 938 K and a characteristic low temperature shoulder at about 700 K. The low temperature peaks were assigned to the reduction of Mo⁶⁺ species to Mo⁴⁺ [34], while the large peak at 938 K was assumed to result from the overlapping of several peaks, corresponding to the reduction of Mo⁴⁺ to Mo⁰ and of P⁵⁺ to P⁰. For the 5 wt% K–10 wt% MoP/SiO₂ catalyst, the peaks shifted to higher temperature compared to the 10 wt% MoP, with peak maxima at 773 and 976 K, and these peak maxima were also present in the 1 wt% Rh–5 wt% K–10 wt% MoP/SiO₂ catalyst. The Rh–K–MoP/SiO₂ catalyst also showed a new reduction peak at about 505 K that was not present in the TPR profiles of the K–MoP/SiO₂ and MoP/SiO₂ catalysts, and was assigned to the reduction of Rh₂O₃ [15,35,36]. From the TPR analysis, there was no clear indication of a strong interaction between the Rh and the MoP since there was no shift in the reduction peak temperatures of the Mo or Rh precursors. The reduction of Rh₂O₃ was assumed to follow the reaction stoichiometry Rh₂O₃ + 3H₂ → 2Rh + H₂O and the stoichiometric H₂ consumption required to reduce the Rh present on the 1 wt% Rh–5 wt% K–10 wt% MoP–SiO₂ catalyst precursor was 145.7 μmol H₂/g. The difference between the total H₂ consumed for the K–MoP/SiO₂ and the Rh–K–MoP/SiO₂ catalyst precursors corresponded to 100% reduction of the Rh₂O₃.

The reduction of Mo and P species from their high oxidation state, Mo⁶⁺ and P⁵⁺ present after calcination, was assumed to follow the reaction 2MoO₃·P₂O₅ + 11H₂ → 2MoP + 11H₂O. Accordingly, the degrees of reduction calculated from Fig. 1 were 78%, 73% and 66% for the 10 wt% MoP/SiO₂ catalyst, the 5 wt% K–10 wt% MoP/SiO₂ catalyst and the 1 wt% Rh–5 wt% K–10 wt% MoP/SiO₂, respectively. However, as noted in several previous studies of supported metal phosphide catalysts [37–40], the stoichiometry of

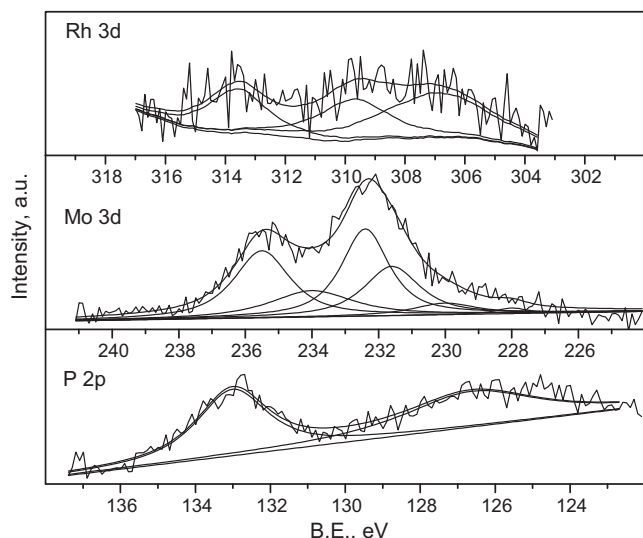


Fig. 2. XPS plots for reduced and passivated Rh-K-MoP/SiO₂ catalyst.

the reduction reaction is somewhat uncertain because of the possible presence of polyphosphate chains [37,38] and species such as $H_xPO_4^{(x-3)}$ and P^{3+} in the calcined precursors. In addition, during calcination and reduction, some loss of P may occur through volatile phosphorous species that leave the catalyst surface during the reduction or calcination process, all of which may contribute to a higher actual degree of reduction than that calculated from the data of Fig. 1 [39].

Table 2 reports the catalyst BET surface areas as well as the catalyst surface composition as determined by XPS analysis of the reduced, passivated catalysts. The BET areas of the catalysts after reaction are also reported in Table 2. The Rh-K-MoP/SiO₂ catalyst BET area (56 m²/g) and the K-MoP/SiO₂ catalyst (64 m²/g) were significantly lower than that of the SiO₂ (330 m²/g) and the 5 wt%K/SiO₂ (264 m²/g) supports. The most significant decrease in area occurred with the addition of MoP to the K/SiO₂ support. The used catalysts all showed a decrease in surface area after reaction, mainly due to carbon deposition [31].

Previous work on K-MoP/SiO₂ catalysts has shown that the Mo:Si ratio, as determined by XPS, is a measure of the Mo dispersion [31], and this is not affected significantly by the addition of the K to the SiO₂ prior to the loading of MoP onto the support. The surface composition data of Table 2 show, however, that with the addition of Rh, there appears to be a loss in MoP dispersion. For the K-MoP/SiO₂ catalyst, the MoP dispersion was 2.8% (i.e. the Mo/Si atom ratio expressed as a percentage) whereas for the Rh-K-MoP/SiO₂ catalyst it was 1.9%. The surface of the catalysts was also enriched in Mo (Mo:P > 1, Table 2), consistent with the bulk chemical analysis, and the Mo enrichment is ascribed to the loss of surface P during thermal treatment of the catalysts.

XPS narrow scan analysis of the reduced and passivated Rh-K-MoP/SiO₂ catalyst is shown in Fig. 2. The Mo3d spectrum of Fig. 2 was de-convoluted using spin orbit splitting of 3.2 eV and three distinct Mo3d_{5/2} binding energies (BEs). The low BE peak (228 eV) was assigned to MoP [39,41] whereas the higher BEs correspond to Mo⁵⁺ and Mo⁶⁺ species that result from the passivation of the catalyst [41]. The P2p spectra of Fig. 2 was de-convoluted into two peaks with BE 133 eV assigned to phosphate species and BE 126.6 eV assigned to phosphide species [41]. Although the Rh narrow scan XPS profile had a relatively high signal-to-noise ratio, a Rh 3d_{5/2} peak at 307.2 eV, corresponding to Rh⁰ species, and a peak at 309.2 eV, corresponding to Rh³⁺, were identified [42,43].

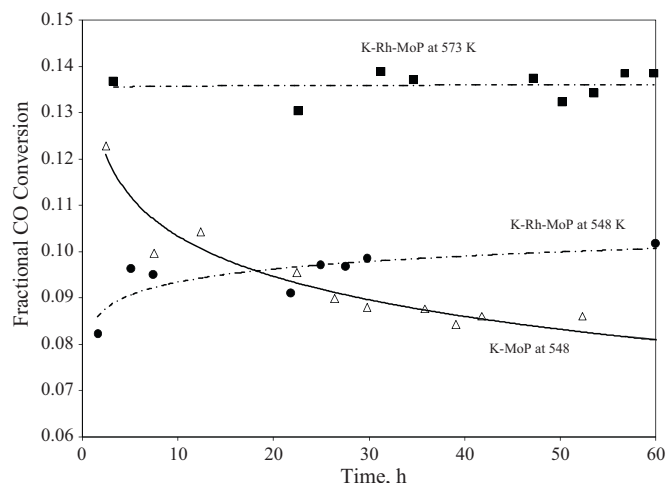


Fig. 3. Comparison between K-MoP and Rh-K-MoP catalyst activity for syngas conversion with time-on-stream at H₂:CO = 1 and 8.27 MPa.

3.2. Comparison of catalysts

Fig. 3 compares the CO conversion measured at 548 K, 8.3 MPa and a H₂:CO ratio of 1 over the 5 wt% K–10 wt% MoP/SiO₂ and the 1 wt% Rh–5 wt% K–10 wt% MoP/SiO₂ catalyst. A sharp decrease in activity was observed for the K-MoP/SiO₂ catalyst at the beginning of the experiment, and similar behaviour was reported previously for a series of catalysts with a range of MoP and K loadings on SiO₂ [31]. However, with the addition of Rh, the activity of the catalysts stabilized, and the same catalyst stability was observed at 573 K and a correspondingly higher CO conversion.

Activity and selectivity data were averaged over the 20–60 h run time period following the initial 20 h stabilization period for the catalysts and these results are summarized in Table 3, comparing the MoP/SiO₂ catalyst product distribution obtained from synthesis gas reacted at 548 K and 8.27 MPa to that from the 5 wt% K–10 wt% MoP/SiO₂ and the 1 wt% Rh–5 wt% K–10 wt% MoP/SiO₂ catalyst. Although there was a significant increase in CO₂ selectivity with K addition to the MoP/SiO₂ catalyst, a smaller increase occurred with addition of Rh to the K-MoP/SiO₂. The addition of K to the MoP/SiO₂ catalyst suppressed the selectivity to CH₄ and increased selectivity to C₂₊ oxygenates. The 5 wt% K–10 wt% MoP/SiO₂ catalyst, operated at 548 K and 8.27 MPa, had a CH₄ selectivity of only 9.7 C atom% and a C₂₊ oxygenate selectivity of 72.6 C atom%. In previous work, the 5 wt% K–10 wt% MoP/SiO₂ catalyst was shown to give the lowest CH₄ and highest C₂ oxygenate selectivity among a series of K-MoP/SiO₂ catalysts [31]. At the same operating conditions, the present work shows that addition of 1 wt% Rh to the K-MoP/SiO₂ catalyst increased the CO conversion from 8.4 to 9.7%. The selectivity to the undesired products, CH₄ and other hydrocarbons, also increased (Table 3). A small decrease in the C₂₊ oxygenate selectivity, increase in ethanol selectivity and a decrease in acetaldehyde selectivity were also observed. The space–time yield (STY) of the C₂₊ oxygenates also decreased marginally. With the Rh-K-MoP/SiO₂ catalyst, the STY for C₂₊ oxygenate was 54.9 g kg cat.^{−1} h^{−1} compared to 59.3 g kg cat.^{−1} h^{−1} for the 5 wt% K–10 wt% MoP/SiO₂ catalyst [30].

Overall, the present data show that addition of Rh to the K-MoP/SiO₂ catalyst improved the stability of the catalyst compared to the K-MoP/SiO₂ catalyst. The Rh increased the selectivity to hydrocarbon products, especially CH₄, while increasing the selectivity to ethanol marginally. The addition of Rh did not appear to significantly increase the hydrogenation of acetaldehyde to ethanol.

Table 2

BET area and surface composition of catalysts as determined by XPS.

Catalyst	MoP/SiO ₂	K-MoP/SiO ₂	Rh-K-MoP/SiO ₂
BET area, m ² /g	263	64	56
BET area after reaction, m ² /g	181	6	24
Surface composition, mole ratio			
Rh:Si	–	–	0.0026
K:Si	–	0.099	0.084
Mo:Si	0.027	0.028	0.019
P:Si	0.024	0.022	0.016
Mo:P	1.10	1.28	1.20

3.3. Activity and selectivity of the Rh-K-MoP/SiO₂ catalyst

The Rh-K-MoP/SiO₂ catalyst was evaluated at different reaction temperatures, in the range of 548–613 K, with H₂:CO feed gas ratios of 1, 1.5 and 2. At each set of operating conditions, the catalyst weight in the reactor was adjusted so as to maintain the CO conversion at approximately the same level and below 25% at each set of conditions. This experimental procedure was followed so that the micro-reactor was operated isothermally and to ensure that the observed kinetics were free of mass and heat transfer resistances. Furthermore, selectivity data reported at a fixed CO conversion are required for proper interpretation. By changing the catalyst weight at each temperature, the average CO conversion was maintained at $17.4 \pm 4.3\%$ for all the data reported herein.

The effect of temperature and H₂:CO ratio on the product selectivity obtained over the Rh-K-MoP/SiO₂ catalyst is presented in Table 4. With increased temperature and fixed H₂/CO ratio, the methane and total hydrocarbon selectivity increased, whereas the C₃₊ oxygenate selectivity decreased. The variation in ethanol and acetaldehyde selectivity was small because the data were collected over a narrow range of CO conversions. The acetaldehyde selectivity was in the range of 18.6–23.3 C atom% and the ethanol selectivity was in the range of 16.0–20.0 C atom%. Methanol selectivity increased with increase in temperature although in all cases the methanol selectivity was below 2 C atom%. Low methanol selectivity is a unique characteristic of the MoP-based catalysts, as reported previously [31]. The CO₂ selectivity showed a decreasing trend with increased temperature and increased H₂/CO ratio. The trend was most pronounced at 598 K. The water-gas-shift (WGS) reaction [CO + H₂O $\leftrightarrow$ CO₂ + H₂, $\Delta H_r = -41.1$ kJ/mol] shifted to the left as H₂ partial pressure and/or temperature increased, leading to a decrease in CO₂ selectivity.

The effect of temperature and H₂:CO ratio on the STY of the individual product components is reported in Table 5. In all cases, the C₂₊ oxygenate STY increased with an increase in temperature. The effect of the H₂/CO ratio was dependent upon the reaction temperature. At 548 K there was a small increase in the C₂₊ oxygenate STY with increased H₂:CO ratio whereas at 613 K, the reverse trend was observed (the C₂₊ oxygenate STY decreased with increased H₂:CO ratio).

3.4. Reaction kinetics

The space-time yield data of the various products generated in the present study, represent the average rate of formation of each component in the reactor. Since the CO conversion levels were kept low as the temperature and H₂:CO ratio varied, the change in H₂ and CO partial pressures through the reactor bed was low (the maximum variation was $\pm 5\%$) for all the data of Table 5. Hence the space-time-yield data can be taken as an estimate of the rate of formation of each of the compounds at the average H₂ and CO partial pressures in the reactor, as reported in Table 5. These data have been used to estimate kinetic parameters for each component using simple empirical power law kinetics of the form $r = ke^{-E/RT} P_{H_2}^m P_{CO}^n$ where k is the reaction rate constant, E is the apparent activation energy and P_{H_2} and P_{CO} are the partial pressure of H₂ and CO raised to a power m and n , respectively. The power law equation was applied to each of the major components as well as the rate of CO consumption (on a CO₂ free basis) and in order to improve the confidence limit on each of the estimated parameters, the hydrogen exponent (m) was assumed to be equal to one. Previous studies that have reported empirical power law kinetics for alcohol synthesis from syngas, report H₂ exponents in the range 0.7–1.3 for methanol, ethanol and propanol [44]. For the range of conditions used in the present study, wherein the H₂ partial pres-

Table 3Effect of promoters on synthesis gas conversion over MoP/SiO₂ catalysts.

Catalyst	MoP/SiO ₂	K-MoP/SiO ₂	Rh-K-MoP/SiO ₂
MoP, wt%	10	10	10
K, wt%	0	5	5
Rh, wt%	0	0	1
Temperature, K	548	548	548
GHSV, h ⁻¹	3960	3960	3960
H ₂ :CO	1.0	1.0	1.0
CO conversion, %	2.8	8.4	9.7
STY C ₂₊ oxygenates, g/(kg cat h)	15.1	59.2	54.9
CO ₂ selectivity, C atom%	24.8	44.9	51.4
Selectivity, C atom% excl. CO ₂			
Hydrocarbons	57.3	23.4	34.9
CH ₄	32.5	9.7	11.7
Methanol	0.5	4.0	0.7
Acetaldehyde	19.1	20.5	18.6
Ethanol	4.4	15.3	16.0
Propanol	2.2	7.4	6.2
Acetone	13.8	14.1	5.3
Total C ₂ + oxygenate	42.6	72.6	64.4

S_{CO₂} = carbon atom% selectivity to carbon dioxide.

Table 4Averaged product selectivities from syngas conversion over the 1%Rh–5%K–10%MoP/SiO₂ catalyst.

Temp, K GHSV, h ^{−1}	H ₂ :CO	% CO conv.	S _{CO₂}	Selectivity (CO ₂ free, C atom%)							
				HC	CH ₄	MeOH	AcH	EtOH	Acetone	C _{2Oxy}	C _{3+Oxy}
548	1.0	9.7	51.4	34.9	11.6	0.7	18.6	16.0	5.3	37.1	27.3
3960	2.0	18.0	44.4	31.8	14.0	0.8	20.5	18.9	9.6	40.9	26.4
573	1.0	13.6	45.6	32.6	13.0	1.2	19.8	19.2	9.0	40.7	25.7
7920	1.5	15.6	44.9	35.4	14.9	1.2	20.8	18.3	10.1	40.7	22.6
	2.0	20.7	43.3	35.8	15.8	1.2	19.6	18.0	9.8	39.2	23.9
598	1.0	13.9	45.0	38.6	16.8	1.5	23.1	20.0	4.5	44.0	16.0
15,840	1.5	17.0	44.5	38.6	17.8	1.3	19.9	18.2	8.6	39.2	20.9
	2.0	21.3	42.7	38.3	18.9	1.2	20.9	18.3	7.6	40.5	20.0
613	1.0	19.7	43.9	40.3	18.2	1.8	21.6	19.5	5.0	41.9	16.0
15,840	2.0	24.4	42.2	42.7	21.1	1.9	23.3	18.8	3.2	42.8	12.7

S_{CO₂} = carbon atom% selectivity to carbon dioxide. HC, total hydrocarbons; AcH, acetaldehyde; ΣC_{2Oxy}, sum of acetaldehyde, ethanol and acetic acid; C_{3+oxy}, acetone, propanol, butanol, propionic acid, methyl acetate, ethyl acetate; NA, data not available.

Table 5Averaged space–time–yields of major products from syngas conversion over the 1%Rh–5%K–10%MoP/SiO₂ catalyst.

Temperature K	Feed H ₂ /CO ratio	Average partial pressure ^a		CO consumption, CO ₂ free μmol g cat ^{−1} s ^{−1}	Component STY					C ₂₊ STY g kg cat ^{−1} h ^{−1}
		H ₂ MPa	CO MPa		CH ₄ μmol g cat ^{−1} s ^{−1}	CH ₃ CHO μmol g cat ^{−1} s ^{−1}	Methanol	Ethanol	Propanol	
548	1.0	4.20	4.09	1.048	0.122	0.097	0.008	0.084	0.022	55
	2.0	5.58	2.64	1.463	0.205	0.150	0.012	0.138	0.025	79
573	1.0	4.22	4.00	3.280	0.426	0.325	0.039	0.315	0.055	174
	1.5	5.04	3.23	3.348	0.499	0.348	0.041	0.306	0.053	169
	2.0	5.58	2.55	3.426	0.541	0.336	0.041	0.308	0.053	174
598	1.0	4.22	3.99	6.777	1.130	0.783	0.100	0.678	0.093	327
	1.5	4.99	3.30	7.363	1.311	0.731	0.093	0.671	0.107	352
	2.0	5.58	2.65	7.144	1.347	0.745	0.083	0.651	0.099	345
613	1.0	4.19	3.92	9.798	1.783	1.058	0.175	0.955	0.143	454
	2.0	5.58	2.64	8.247	1.740	0.961	0.157	0.755	0.109	368

^a Average of inlet and outlet H₂ and CO partial pressures.

sure varies over a relatively narrow range (from 4.2 to 5.6 MPa, Table 5) and for which the H₂ and CO partial pressures were highly correlated, an assumed exponent of $m = 1.0$ suffices. The unknown parameters k , E and n were evaluated by fitting the power law equation to the experimental STY data. The Marquardt–Levenberg optimization procedure was used to evaluate the parameters by minimizing the objective function $S = \sum_{i=1}^N (r - r_{\text{mod}})^2$ where r_{exp} is the measured STY of the components of interest and r_{mod} is the STY calculated from the power law model. The kinetic parameters for the four major components are reported in Table 6 and the parity plot, shown in Fig. 4, confirmed the goodness-of-fit between the experimental observations and the model calculations. The data of

Table 6 show that the apparent activation energy of the alcohols decreased as the alcohol carbon number increased. Furthermore, the activation barriers associated with the formation of acetaldehyde and ethanol were approximately equal (within the standard deviation associated with the parameter estimates), and these were both significantly lower than the apparent activation energy for methanol.

4. Discussion

The catalyst characterization data show that addition of Rh to the K–MoP/SiO₂ catalyst decreased the catalyst BET surface area

Table 6Power law rate parameters for major products over the 1%Rh–5%K–10%MoP/SiO₂ catalyst.

Component	Power law rate expression, $r = A \exp\left(-\frac{E}{RT}\right) P_{\text{H}_2} P_{\text{CO}}^m$, μmol g cat ^{−1} s ^{−1}			
	Activation energy, E kJ/mol	Power law exponent, m –	Pre-exponential factor, A (μmol) (MPa) ^{−(m+1)} (s) ^{−1}	R^2 %
CO consumption ^a	75.0 ± 5.9 ^b	0.74 ± 0.14	0.3898	96.1
Methane	91.3 ± 6.7	0.48 ± 0.15	0.0871	96.8
Acetaldehyde	81.9 ± 5.1	0.77 ± 0.12	0.0388	97.5
Methanol	114.5 ± 10.9	1.00 ± 0.21	0.0033	94.2
Ethanol	77.7 ± 5.1	0.93 ± 0.12	0.0288	97.1
Propanol	66.9 ± 9.9	0.79 ± 0.25	0.0055	86.3

^a CO₂ free basis.

^b Std. deviation of parameter estimate.

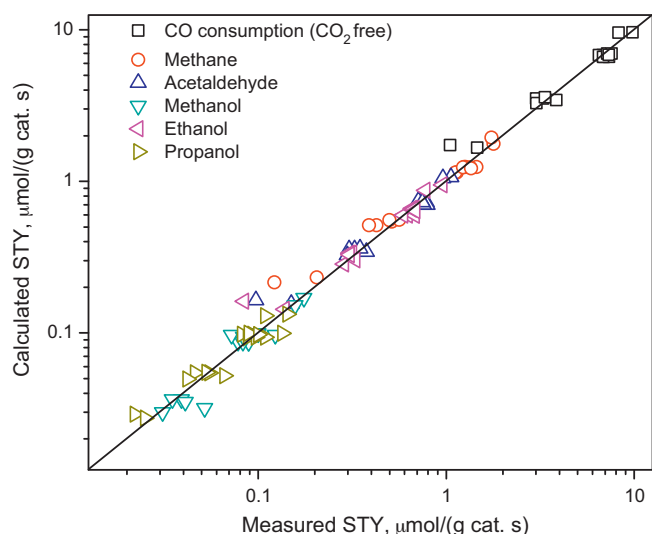


Fig. 4. Parity plot for power law kinetic models shown in Table 6.

and the MoP dispersion. This suggests that the Rh was dispersed over the support as well as the MoP, although the MoP surface concentration on the support was about $10\times$'s greater than that of Rh. The TPR profile of the MoP precursors was not affected by the addition of Rh and the TPR data showed that 100% of the Rh precursor was reduced to Rh metal. In addition, the BEs of the Mo and P species on the passivated Rh–K–MoP/SiO₂ catalyst were similar (within the BE resolution capability of the instrument) to those reported previously for K–MoP/SiO₂ catalysts [31]. Hence, the catalyst characterization data suggest that the Rh was dispersed over the catalyst as Rh⁰ and that the presence of Rh did not influence the properties of the MoP.

In previous work, it was shown that on K–MoP/SiO₂ catalysts, both acetaldehyde and ethanol were the major C₂ oxygenated products of syngas conversion [31]. The addition of Rh to the K–MoP/SiO₂ catalyst was expected to enhance ethanol selectivity by hydrogenating the acetaldehyde to ethanol [45]. However, the data of Table 3 showed that this did not occur to any great extent. Addition of Rh increased the selectivity to CH₄ and other hydrocarbons and, to a much lesser extent, increased the selectivity to ethanol, although a small decrease in the total C₂+ oxygenate selectivity was observed.

Several Mo-based catalysts have been investigated for CO hydrogenation to alcohols and hydrocarbons, and Table 7 summarizes some recently reported results. Table 7 shows that in the absence of alkali metal oxides, Mo, Mo₂C and MoS₂ yield mostly methane and other hydrocarbons, similar to that obtained on the MoP/SiO₂ catalyst of the present study (Table 3). Although addi-

tion of K to Mo₂C and MoS₂ catalysts decreased CO conversion, the selectivity to alcohols and other oxygenates increased while hydrocarbon selectivity was suppressed [46,47]. The data of Table 3 show similar effects with K addition to the MoP/SiO₂ catalyst, as reported previously [31].

The effect of addition of Group VIII metal promoters to the Mo-based catalysts is somewhat more complicated. Table 7 shows that the addition of Co to a K–MoS₂ catalyst decreased hydrocarbon selectivity while increasing CO conversion, C₂+ oxygenate selectivity and alcohol STY, and similar data have been reported by others [20–22]. An increased alcohol selectivity and alcohol STY was also observed with Rh addition to a K–Mo₂S catalyst [23]. However, addition of Co to the K–Mo₂C catalyst increased CO conversion and alcohol STY but, significantly, the hydrocarbon selectivity also increased. The K–Co–Mo catalyst [49] also had high selectivity to hydrocarbons. In the present work, increased hydrocarbon selectivity was also observed with Rh addition to the K–MoP/SiO₂ catalyst. Together, these results show that the Group VIII metal added to Mo, Mo₂C and MoP catalysts behave differently compared to when they are added to the MoS₂ catalysts. In the latter case, especially when the catalysts are pre-treated in H₂S, the Group VIII metals are readily sulphided [47], whereas in the case of Mo, Mo₂C and MoP, there is no exposure to S. In the case of Rh, Koizumi et al. [47] has shown that sulphided Rh species (Rh₁₇S₁₅) have high methanol selectivity and overall alcohol selectivity significantly greater than the hydrocarbon selectivity. Similarly, when MoS₂ is promoted with Co, the surface species include a Co₈S₉ species [54] and the K–Co–MoS₂ catalyst also has increased yield of alcohols compared to K–MoS₂ catalysts. Hence we conclude that the effect of Group VIII metal promotion of MoS₂ catalysts results in at least partially sulphided species whereas in the case of Mo, Mo₂C and the MoP of the present work, the Co and Rh remain in a metallic state, and for this reason the hydrocarbon selectivity increases on these catalysts.

The hydrogenation capabilities of the Rh present in the Rh–K–MoP/SiO₂ catalyst is also likely responsible for the improved stability of the catalyst compared to the K–MoP/SiO₂ catalyst. As noted previously [31], the deactivation of these catalysts is linked to carbon deposition from heavy products or carbon generated by the Boudouard reaction. In either case, improved hydrogenation of the catalyst would be expected to reduce the carbon deposition by hydrogenation to methane or other light hydrocarbons, and thereby improve catalyst stability.

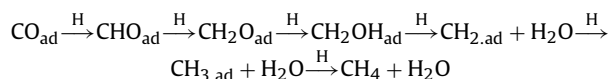
The MoP catalysts of the present study show promise for several reasons. Firstly, MoP has higher selectivity to ethanol compared to Mo₂C, MoP has higher C₂+ oxygenate selectivity than other Mo-based catalysts (Tables 3 and 7) and the products are not contaminated with S as in the case of MoS₂ [44]. The MoP catalysts also have relatively low hydrocarbon selectivity. In addition, in most other alcohol synthesis catalysts, (MoS₂, Mo₂C, Cs–Cu–ZnO, modified FT catalysts) methanol is the major or second major liquid

Table 7
Effect of promoters on Mo-based catalysts for synthesis gas conversion.

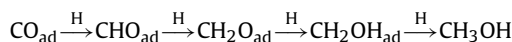
Catalyst Promoters	MoS ₂				Mo ₂ C			Mo	
	–	K	K–Co	K–Rh	–	K	K–Co	–	K–Co
Reference	[55]	[48]	[48]	[23]	[47]	[47]	[56]	[49]	[49]
Temperature, K	613	583	583	600	573	573	573	503	503
GHSV, h ^{−1}	6000	4800	4800	4800	2000	2000	2000	4800	4800
H ₂ :CO	2.0	2.0	2.0	2.0	1.0	1.0	1	2.0	2.0
Pressure, MPa	4.1	5.0	5.0	10.0	8.0	8.0	8	5.0	5.0
CO conversion, %	50	5	15	11	59	23	51	0.3	12
CO ₂ selectivity, C atom%	42	20	26	–	52	50	–	–	–
Selectivity, C atom% excl. CO ₂									
Hydrocarbons	100	57	16	41	96	47	60	100	51
Methanol	0	24	26	11	3	19	11	0	6
Total C ₂ + oxygenate	0	20	58	48	1	34	29	0	43
Alcohol STY, g/(kg cat h)	0	31	189	174	16	122	161	0	185

product [1,2], whereas for the MoP catalyst, methanol selectivity was the lowest among the C₁–C₃ oxygenated products. A low selectivity to methanol also occurs on Rh-based catalysts [12,14,15,45].

On MoP and MoS₂ catalysts, CO is adsorbed non-dissociatively [29]. DFT studies of an Mo₆P₃ cluster model of MoP [50] identified the reaction mechanism for synthesis gas conversion to methane as:



and for methanol synthesis as:



Similar mechanisms involving the same surface intermediates have also been reported for MoS₂ catalysts [51]. On MoP the C–O bond breaks upon H_{ad} addition to the O of the hydroxymethylene (CH₂OH_{ad}) species, forming CH_{2,ad} and H₂O_{ad}. The CH₂ species participate in CO insertion reactions or other C–C bond forming reactions with other surface species to yield higher oxygenates [45]. DFT studies on the Mo₆P₃ cluster model of MoP identified the formation of methanol from hydroxymethylene (CH₂OH) or methoxy (CH₃O) adsorbed species as having a much higher activation barrier than C–O bond breakage of hydroxymethylene to produce the CH₂ species. Accordingly, the activation barrier for C₂+ oxygenated products would be expected to be less than that for methanol formation. The apparent activation energies reported in Table 6 support these modeling results in that they show a higher apparent activation energy for methanol formation than for either ethanol, acetaldehyde or propanol formation.

The apparent activation energies estimated for ethanol and acetaldehyde formation on the Rh–K–MoP/SiO₂ catalyst were very similar (Table 6), suggesting that a common intermediate is involved in the kinetic steps leading to these products. Most kinetic models of ethanol synthesis on Rh favour a mechanism in which ethanol is produced by the secondary hydrogenation of acetaldehyde, and assume that ethanol and acetaldehyde are formed through a common pathway [1,45]. For the present catalyst, however, acetaldehyde was not hydrogenated to ethanol as the H₂:CO ratio increased and both acetaldehyde and ethanol were significant products over the K–MoP/SiO₂ and the Rh–K–MoP/SiO₂ catalysts. The MoP does not show the hydrogenation ability to convert acetaldehyde to ethanol that is apparent on Rh catalysts.

On Rh-based catalysts the reaction begins with CO dissociation and hydrogenation to yield CH_x species which then undergo hydrogenation to CH₄ as well as CO insertion and hydrogenation to yield higher alcohols and hydrocarbons [45]. The increase in hydrocarbon selectivity observed upon Rh addition to the K–MoP/SiO₂ catalyst suggests that Rh hydrogenated adsorbed CH₂ species to produce methane or higher hydrocarbons and these species likely resided on the Rh, but may also have originated from the MoP. However, the relatively low selectivity to hydrocarbons obtained over the K–MoP/SiO₂ catalyst suggests that there are relatively few of these species on the MoP catalyst. On the other hand, the fact that the acetaldehyde selectivity decreased and ethanol selectivity increased only marginally with addition of Rh to the K–MoP/SiO₂ catalysts, is indicative of oxygenated surface species, mostly on the K–MoP catalyst that do not migrate to Rh, and hence do not undergo further hydrogenation.

The activation energy for methane formation on a Rh–CeO₂/SiO₂ catalyst was reported as 92.5 kJ/mol [15], for a 1%Rh/SiO₂ catalyst [14] a value of 94.6 kJ/mol has been reported and on a 1%Rh/ZrO₂ catalyst a value is 134.6 kJ/mol has been reported [52,53]. The Rh–K–MoP/SiO₂ catalyst of the present study had activation energy of 91.3 kJ/mol for methane formation. The relatively similar values of the activation energies for methane formation suggest

that on the Rh–K–MoP/SiO₂ catalyst, a significant fraction of the methane formed on the Rh surface. For the same series of catalysts, the activation energy for ethanol formation was reported as 49.4 kJ/mol on Rh–CeO₂/SiO₂ and 66.5 kJ/mol on 1%Rh/ZrO₂. On the Rh–K–MoP/SiO₂ catalyst the activation energy is significantly higher at 77.7 kJ/mol, suggesting that ethanol and other oxygenates are mainly formed on the K–MoP species of this catalyst, rather than on the Rh.

5. Conclusions

The MoP catalysts showed a distinct product distribution compared to other Mo-based catalysts for syngas conversion. Low selectivity to methanol and moderate selectivity to a mixture of C₂ oxygenates (acetaldehyde, ethanol and acetic acid) was achieved on the K–MoP–SiO₂ and the Rh–K–MoP/SiO₂ catalysts. Addition of Rh to the K–MoP/SiO₂ catalyst improved the stability of the catalyst compared to the K–MoP/SiO₂ catalyst. However, the Rh increased the selectivity to hydrocarbon products, especially CH₄, while only increasing the selectivity to ethanol marginally. The results suggest that the oxygenated products were mainly produced on the K–MoP surface. The empirical power law kinetic parameters suggested a common reaction intermediate for ethanol and acetaldehyde. Hydrogenation of acetaldehyde to ethanol did not occur to any great extent, even in the presence of the Rh.

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